

LLD Master Handbook

Based on our complete learning journey and interview practice sessions

Chapter 1 - Core Mindset

- LLD is not about memorizing classes.
- First understand entities, actions, data and relationships.
- Interviewers check thinking, not diagrams alone.

Chapter 2 - Golden Rules

- Noun -> Class Candidate
- Verb -> Method Candidate
- Simple Value -> Variable
- IS-A -> Inheritance
- HAS-A -> Composition/Aggregation
- Controller Class -> Coordinates everything

Chapter 3 - Library Practice

- Book = Class
- User = Class
- Library = Class
- IssueBook = Method
- ReturnBook = Method
- SearchBook = Method
- Lesson: Actions should not become classes.

Chapter 4 - School Management Practice

- Student = Class
- Teacher = Class
- Course = Class
- registerStudent() = Method
- Lesson: Nouns become entities, actions become methods.

Chapter 5 - Food Delivery Practice

- User, Restaurant, DeliveryPartner, Menu, Order, Payment = Classes
- placeOrder(), makePayment(), deliverOrder() = Methods
- Restaurant HAS-A Menu
- Order HAS-A Payment

Chapter 6 - ATM Practice

- ATM, Customer, BankAccount, Card, Transaction = Classes
- withdraw(), checkBalance() = Methods
- PIN, Balance = Variables
- Controller Class = ATM

Chapter 7 - Movie Booking Practice

- Movie, Theatre, Show, Seat, Ticket, Payment = Classes
- searchMovie(), bookTicket(), selectSeat() = Methods
- Controller = BookMyShowSystem

Chapter 8 - WhatsApp Practice

- User, Chat, Message, Group = Classes
- sendMessage(), receiveMessage() = Methods
- phoneNumber, lastSeen = Variables
- Admin IS-A User

Chapter 9 - Location, Rating and Trip

- Location = Class (city, state, zip etc.)
- Trip = Class (source, destination, fare, status)
- Rating = Variable in simple systems
- Rating = Class in advanced systems

Chapter 10 - IS-A vs HAS-A

- Dog IS-A Animal
- Admin IS-A User
- Bike IS-A Vehicle
- Car HAS-A Engine
- Library HAS-A Books
- Trip HAS-A Location

Chapter 11 - Parking Lot Design

- Vehicle = Parent Class
- Car, Bike, Truck = Child Classes
- ParkingSpot, Ticket, Payment = Classes
- entryVehicle(), exitVehicle(), calculateFee() = Methods
- ParkingLot = Controller Class
- ParkingLot HAS-A ParkingSpots

Chapter 12 - Parking Spot Inheritance

- CompactSpot IS-A ParkingSpot
- LargeSpot IS-A ParkingSpot
- BikeSpot IS-A ParkingSpot

Chapter 13 - Library Deep Dive

- Book = Class
- BookCopy = Class
- Author = Class
- BookTitle = Variable
- ISBN = Variable
- Key Insight: Library issues BookCopy, not Book

Chapter 14 - Chess Design

- ChessGame = Controller
- Player, Board, Piece = Classes
- King, Queen, Rook, Bishop, Knight, Pawn inherit Piece
- movePiece(), checkWinner() = Methods
- Position = Class

Chapter 15 - How To Decide Class vs Variable

- Ask: Does it have identity?
- Does it have multiple attributes?
- Can it have future behavior?
- If yes -> Class
- If only value -> Variable

Chapter 16 - Controller Class Identification

- ATM -> ATM
- Parking Lot -> ParkingLot
- Library -> Library
- Chess -> ChessGame
- Food Delivery -> OrderManager/FoodDeliverySystem

Chapter 17 - High Priority Design Patterns

- 1. Strategy Pattern
- 2. Factory Pattern
- 3. Observer Pattern
- 4. Decorator Pattern

- 5. Singleton Pattern
- 6. Adapter Pattern
- 7. Builder Pattern
- 8. State Pattern
- 9. Command Pattern
- 10. Chain of Responsibility

Chapter 18 - Interview Cheat Sheet

- Extract Nouns
- Extract Verbs
- Identify Variables
- Find Relationships
- Find Controller
- Think Extensibility
- Apply SOLID Principles

LinkedIn Summary

This handbook captures a practical journey from beginner entity identification to interview-oriented LLD thinking. The focus is on identifying classes, methods, variables, inheritance, composition and controller classes using real interview-style examples.